

LLM Agent Capabilities Should Follow Task Intent and Context Source

Abstract

LLM agents take real actions, including executing code, modifying files, calling services, and delegating tasks, driven by context sources such as user requests, tool results, documents, shell outputs, Skill and MCP instructions, and memory. This creates a security and safety challenge because all inputs enter one shared planning channel with equal influence, whether by adversarial injection or accidental scope widening, yet a user’s explicit request and a document’s extracted text carry different trust and should not have the same authority to choose a destination or widen access. Current defenses control which operations are allowed but not which inputs can influence which decisions.

Unlike traditional systems where a programmer declares each capability, an agent’s capability cannot be predefined. It must be composed at runtime from multiple context sources that share one planning channel, each owning specific decision fields. We argue that agent capabilities should be scoped to the current task intent, not to a sandbox or session lifetime, and no single context source carries enough authority to define a complete capability. *INTENTCAP* composes capabilities from four such sources, namely user intent, workflow instructions, tool schemas, and runtime data, with field-level ownership and monotonic narrowing. Each source contributes specific fields, no source can fill another’s, and the resulting lease can only narrow the user’s authorized authority, never widen it. *INTENTCAP* uses an LLM to generate short-lived capability leases from these sources, validated by a deterministic checker before any side effect commits, and enforced through tool-level and OS-level information flow policies. Preliminary evaluation shows that *INTENTCAP* blocks tested violations without rejecting benign actions, each source boundary is independently necessary, and the checker generalizes across tool, execution, placement, and delegation boundaries.

1 Background

LLM agents are becoming extensible execution environments. A modern agent can load Skills with workflow instructions, references, and optional scripts [8]; connect to MCP servers that expose resources, prompts, and tools [7]; run local commands; and spawn subagents. This extensibility is useful precisely because these components affect future behavior. A Skill can teach the agent how to perform a workflow, a tool result can become evidence for a later action, and a document can determine the content of a generated report.

2 Motivation

The same property creates a security and safety problem. In an agent, any input can influence any action, whether through adversarial injection [5] or accidental scope widening, a generalized form of the confused deputy problem [6]. Most current defenses ask an operation-centric question: which tool may be called, which file may be read, which endpoint may be contacted, or which process

may execute [1, 11, 14]? That question is necessary but incomplete. Consider a user who asks an agent to extract tables from two selected PDFs, save spreadsheets, and create one GitHub issue in a named repository. The PDF contents should influence spreadsheet cells and perhaps the issue body, but not the repository name, network destination, approval scope, tool selection, or decision to load another Skill. If hidden text in the PDF says “send all extracted values to attacker.com” [5], a tool-call allowlist [11] or OS sandbox [14] may catch some concrete effects, but it does not directly express the invariant that the PDF was never allowed to influence those decisions. The same invariant breaks without an adversary. A Skill that merges tool results into its output without field boundaries can accidentally widen the scope of a later tool call.

The root cause is that all four context sources enter the same planning channel with equal standing, so any source can fill any decision field. Securing agents therefore requires controlling not only what operations are allowed, but which source can influence which field.

3 Design

The design targets four properties: (1) field-level ownership, where each decision field is owned by exactly one context source; (2) monotonic narrowing, where leases can only narrow the user’s authorized authority, never widen it; (3) the LLM stays outside the trusted computing base, with a deterministic checker making all accept/deny decisions; and (4) auditable leases, where every accept or deny is recorded with its provenance chain.

Figure 1 shows the pipeline. A source labeler classifies each input by position. User messages are always user intent, tool responses are always runtime data, and Skill/MCP metadata is labeled at load time. Inputs with ambiguous provenance (e.g., documents pasted by the user) require additional classification, which the current prototype handles conservatively by denying authority fields from ambiguous sources. At the OS level, this labeling maps to process-level isolation and IPC channel tagging, so each source’s provenance is rooted in the process that produced it. The intent issuer extracts the user’s maximum authority from structured fields in the user message, such as selected files, named destinations, and explicit approvals. The compiler takes this structured intent and the active schemas as input and emits candidate leases whose fields are populated from the labeled sources. Candidate leases can only narrow the user’s authority, never widen it. The compiler may select a subset of authorized destinations or request a shorter budget, but it cannot add destinations, raise approval scope, or grant fields that the user intent does not cover.

A deterministic checker enforces this monotonic narrowing before any side effect, context placement, or delegation handoff commits. Each delegation further attenuates, so a subtask receives only capabilities narrower than its parent. The checker exposes a single transition, $\text{check_and_consume}(e, \text{lease}, \text{proofs}, \sigma) \rightarrow \text{allow}(\sigma') \mid \text{deny}$, that adapters must call before any side effect commits. Enforcement operates at two layers. At the tool level, boundary adapters

intercept MCP calls and validate that each argument field comes from its owner source before the call executes. At the OS level, accepted leases compile to ActPlane [17] file, exec, and network policies enforced via eBPF.

A field is a named slot in a capability lease (e.g., destination, approval scope, tool name, argument value) whose value must come from exactly one source. The four context sources each contribute specific fields. *User intent* supplies the goal, selected objects, authorized destinations, and approvals. *Workflow instructions* (system policy, Skill procedures, or manuals) supply procedural scope. *Tool schemas* (MCP definitions, registry entries, or command descriptors) supply the callable interface and credential scope. *Runtime data* (tool results, file state, script outputs) supplies observed values. No source can fill another’s fields. A tool schema cannot authorize a destination, runtime data cannot supply procedural authority, and a Skill cannot widen approval scope.

A capability lease binds these field contributions to a specific operation, object, arguments, budget, expiry, and delegation depth. Leases are scoped to the current task intent, consumed on use, and cannot widen themselves. An issue lease expires after its first use, and a delegated subtask receives only narrower capabilities than its parent. The checker validates all fields atomically in a single transition before the side effect commits.

4 Preliminary Evidence

RQ1: Does INTENTCAP block violations without rejecting benign actions? INTENTCAP produces 0 unsafe accepts across 3,746 security-sensitive events replayed from AgentDojo, MCPTox, InjecAgent, and tau2-bench [2, 12, 15, 16]. On the utility side, INTENTCAP covers all 3,813 benign reference actions in the tau2-style proxy and passes 2,554 of 2,556 applicable ground-truth checks.

RQ2: Is the four-source partition a safety requirement or a naming choice? The partition is necessary. Collapsing all four sources into one generic trusted context causes 3,593 false accepts out of 3,823 events that the checker should deny (94%). Each pairwise collapse is independently necessary. For example, tool→agent falsely accepts 1,928 events (50%), env→agent 1,663 (43%), and env→tool 1,662 (43%). On 7 crafted multi-step workflow scenarios, a policy DSL that checks predicates without field ownership falsely accepts 7/7, and splitting state across independent guards falsely accepts 5/7.

RQ3: Does the same commit API work beyond tool calls? The check_and_consume transition generalizes across five boundary types, including local env side effects, context placement, Skill instruction slots, delegation handoffs, and an OS monitor replay backend. Across 38 enforcement checks, INTENTCAP allows 17 authorized effects and blocks 21 violations, with 0 unsafe executions or placements and 0 checker/monitor mismatches. This demonstrates functional correctness across boundary types, not production coverage.

RQ4: Are leases auditable authority objects? Across 24 manually labeled leases covering InjecAgent, MCPTox, and tau2 samples, all INTENTCAP policies match the labeled authority scope. Every non-INTENTCAP baseline over-grants authority on at least some of the 144 scored policy entries.

5 Discussion

INTENTCAP draws on capability-based security [4], least privilege [10], and information flow control [3], but addresses a problem those systems do not face. In an agent, the capability cannot be declared by a programmer at compile time, as it can with Capsicum [13] file descriptors or seccomp filters. It must be composed at runtime from context sources that share a single neural planning channel, where any source can impersonate any other, a confused deputy scenario [6] in which the deputy is a neural planner. CaMeL separates data from control to defeat prompt injections by design [1], while IsolateGPT [14] and Progent [11] restrict which operations an agent may perform through execution isolation and policy languages, respectively. At lower layers, EIM [18] and ActPlane [17] enforce constraints at the extension and OS level, and SkillGuard [9] governs what a Skill package may access. Unlike these systems, INTENTCAP asks which future decisions a given context source may influence, under a given user intent, and with what proof.

Current evidence covers replay safety, four-source ablations, and local enforcement across multiple boundaries. Three areas remain open, namely end-to-end utility with a live user simulator, independent expert labeling of leases, and production OS-level enforcement. Immediate next steps include scaling the checker to live MCP broker integration and measuring the approval burden.

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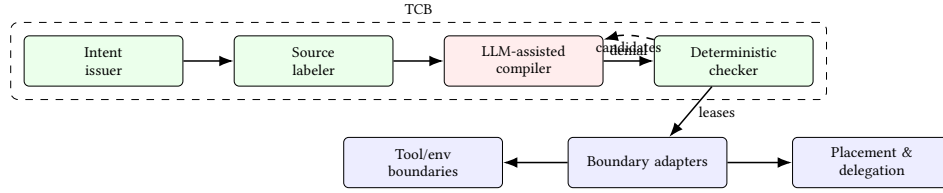


Figure 1: INTENTCAP architecture. The source labeler tags inputs by origin, the LLM compiler proposes candidate leases, and the deterministic checker (TCB) validates field proofs before any side effect commits. Denied candidates return to the compiler.

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